

IBHAUDUR RAHMAN

Full Stack AI Engineer

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PROFESSIONAL SUMMARY

Full Stack AI Engineer with 5+ years of experience building and maintaining enterprise web applications and AI-powered solutions. Strong experience with **React.js, Next.js, TypeScript, Node.js, Python, and FastAPI**, with hands-on experience across frontend, backend, database, cloud, and deployment layers. Experienced with **PostgreSQL, MongoDB, pgvector, REST APIs, Docker, AWS, CI/CD, and Nginx**. Hands-on experience building Generative AI solutions using **LLMs, Retrieval-Augmented Generation (RAG), embeddings, vector search, LangChain, LangGraph, and Model Context Protocol (MCP)**. Strong understanding of **system design, application security, authentication, authorization, rate limiting, concurrency, caching, and performance optimization**.

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, TanStack Query, Material UI, Tailwind CSS, Mantine UI.

Backend: Node.js, Express.js, Python, FastAPI, REST APIs, GraphQL, Strawberry, API Design, Authentication, Authorization, Rate Limiting, Concurrency

Databases: PostgreSQL, MongoDB, pgvector, SQL, Database Design, Indexing, Query Optimization

AI / Generative AI: LLMs, RAG, LangChain, LangGraph, MCP, Embeddings, Vector Search, Semantic Search, Prompt Engineering, AI Agents, Ollama

Cloud & DevOps: AWS, Docker, Docker Compose, Nginx, CI/CD, Linux, Git, GitHub

Architecture & Security: System Design, Scalable Architecture, Caching, Performance Optimization, Application Security, AES Encryption, Secure Data Handling

EXPERIENCE

Software Engineer II

Bangalore, Karnataka

Bayer

Dec 2025 - Present

Projects: Workforce Wizard | LiveLine

- **Continued working on Workforce Wizard after transitioning from HCLTech to Bayer**, contributing to an enterprise application supporting **40K+ users**, delivering **13+ features** across **23 production releases** using **React.js, Next.js, TypeScript, Python, FastAPI, and PostgreSQL**.
- Improved application performance using **Redis caching, lazy loading, and TanStack Query**, reducing redundant API/database calls and improving cached data retrieval.
- Worked across **Docker, AWS, CI/CD, and Nginx** for application deployment and production environments.
- Contributed to **LiveLine**, an in-house solution replacing externally outsourced software and delivering **significant cost savings for Bayer**, using **GraphQL and Strawberry** for API development.
- Built **2 Generative AI use cases** using **LLMs, RAG, embeddings, LangChain, and LangGraph**, with RAG across **10 data sources** and **1 MCP integration**.
- Automated employee shift-scheduling workflows, reducing manual Excel-based effort by approximately **90%**.
- Applied **security, concurrency, caching, performance optimization, and scalable application practices** across production systems.

Technologies: React.js, Next.js, TypeScript, Python, FastAPI, GraphQL, Strawberry, PostgreSQL, Redis, TanStack Query, Docker, AWS, CI/CD, Nginx, LLMs, RAG, LangChain, LangGraph, MCP

Lead Engineer

Bangalore, Karnataka

HCLTech

Mar 2025 - Nov 2025

Project: Workforce Wizard

- Developed **8 features** and approximately **30+ REST APIs** using **React.js, TypeScript, Python, and PostgreSQL** for an enterprise application.
- Improved application responsiveness using **caching** and server-side processing to reduce client-side memory usage and computational overhead.
- Containerized services with **Docker** and supported deployments using **AWS, CI/CD, and Nginx**, contributing to approximately **10 deployments**.

- Improved maintainability through **reusable components/functions, modular architecture, and TypeScript-based type safety**.
- Troubleshoot issues across **frontend, backend, database, and deployment layers** while collaborating with developers and QA.

Technologies: React.js, TypeScript, Python, PostgreSQL, Docker, AWS, CI/CD, Nginx

Software Developer

Chennai, Tamil Nadu

KIRSHI TECHNOLOGIES AND CONSULTING Pvt. Ltd.

Feb 2022 - Feb 2025

Projects: Super Admin Portals | Merchant Portals | Web Applications | 14+ Client Projects

- Developed and maintained **14+ web applications and portals**, including **Super Admin and Merchant portals**, using **React.js, Next.js, Redux, Material UI, and Tailwind CSS**.
- Improved React/Redux application performance by reducing unnecessary renders by approximately **40%** and optimized data-fetching workflows, reducing response/loading time by approximately **25%**.
- Built **reusable components and modular UI patterns**, improving code reusability and development efficiency by approximately **50%** across multiple projects.
- Developed and integrated **RESTful CRUD APIs** using **Node.js, Express.js, and MongoDB**, connecting backend services with React/Next.js applications.
- Implemented **AES encryption** for secure client-server data transmission and handling of sensitive application data.
- Collaborated across multiple projects to develop new features, troubleshoot production issues, optimize application performance, and deliver **scalable and maintainable web solutions**.
- Worked with **Git** for version control and followed modular development practices across multiple client applications.

Technologies: React.js, Next.js, Redux, React Query, Node.js, Express.js, MongoDB, Material UI, Tailwind CSS, Git

Internship

Kumbakonam, Tamil Nadu

Vebbox Software Solution

Sep 2021 - Feb 2022

Learned and applied **Fullstack development** in application.

- Created clones of **YouTube, WhatsApp**, and a **watch website** using React.js, emphasizing intuitive user interaction and responsive design.
- Enhanced user-friendly interfaces and reliable systems in **MERN** environments.

PERSONAL PROJECTS

AI-Powered Portfolio Assistant | RAG Application

- Built an **AI-powered assistant** using **Retrieval-Augmented Generation (RAG)** to answer questions from portfolio and technical documents.
- Developed the AI service using **Python and FastAPI**.
- Used **LangChain** for document processing, retrieval, and LLM orchestration.
- Generated **embeddings** and stored vectors in **PostgreSQL with pgvector**.
- Implemented **semantic similarity search** to retrieve relevant context before generating responses.
- Integrated **Ollama and local LLMs** for response generation.
- Designed separate **frontend, API, AI service, and database layers**.
- Containerized services using **Docker and Docker Compose**.

Technologies: Next.js, TypeScript, Python, FastAPI, PostgreSQL, pgvector, LangChain, Ollama, Docker, RAG, LLMs

EDUCATION

Master of Computer Application (MCA)

Chennai

Dr MGR Educational and Research Institute

2021 - 2023

Bachelor of Computer Application (BCA)

Trichy

Jamal Mohamed College

2017 - 2020